The Principle of Inclusion and Exclusion: A Deeper Connection to Pascal's Triangle

Sophia DeGregorio

1 Introduction

Whether we realize it or not, we have all used the Principle of Inclusion and Exclusion (PIE) before. In its simplest form, it tells us that the number of items in 2 groups is the sum of the number of items in each group minus the number of items in both groups. For example, we would use it to find the number of students in at least 1 of a math or science class, given that there are 34 students taking math, 23 taking science, and 16 taking both: there are $34 + 23 - 16 = 41$ students in at least 1 of the classes. More formally, this formula can be written as

$$|A \cup B| = |A| + |B| - |A \cap B|$$

for 2 sets A and B, where $|X|$ denotes the cardinality of a set X, or the number of elements of X.

But what if we have more than 2 sets? Or what if we want to know the number of elements in, say, at least 2 of the sets? At least 3? This first question is answered by a widely known formula for PIE for n sets. The answer to the others, however, is less well-known, and it turns out it is closely related to Pascal's Triangle.

2 PIE for Counting the Number of Items in the Union of n Sets

For n sets $A_1, A_2, \ldots, A_n$, PIE states that

$$\begin{aligned}
&|A_1 \cup A_2 \cup \cdots \cup A_n| \\
=&|A_1| + |A_2| + \cdots + |A_n| \\
&- (|A_1 \cap A_2| + |A_1 \cap A_3| + \cdots + |A_{n-1} \cap A_n|) \\
&+ |A_1 \cap A_2 \cap A_3| + \cdots + |A_{n-2} \cap A_{n-1} \cap A_n| \\
&- (|A_1 \cap A_2 \cap A_3 \cap A_4| + \cdots + |A_{n-3} \cap A_{n-2} \cap A_{n-1} \cap A_n|) \\
&\vdots \\
&+ (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|.
\end{aligned}$$

We can write this more succinctly by letting

$$S_k = \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}|.$$

PIE then states that for $A_1, A_2, \ldots, A_n$,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} (-1)^{i-1} S_i.$$

2.1 Proof of PIE

To prove that this formula accurately counts the cardinality of the union of the n sets $A_1, A_2, \ldots, A_n$, we must ensure that each element in this union is counted exactly once. Take an arbitrary element x that is in m of the n sets for some $1 \leq m \leq n$. We then consider the value of each of the lines in the multiline expression for PIE:

1. x is in $\binom{m}{1}$ of the sets.
2. x is in $\binom{m}{2}$ of the intersections of 2 sets.
3. x is in $\binom{m}{3}$ of the intersections of 3 sets.

   $\vdots$

m. x is in $\binom{m}{m}$ of the intersections of m sets, namely the intersection of the m sets it is in, on the mth line. After this line, x is counted 0 times because it does not appear in any of the subsequent intersections.

We thus aim to show

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m-1}\binom{m}{m} = 1.$$

Notice that the left-hand side of this equation is equal to

$$-\binom{m}{0} + \binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m-1}\binom{m}{m} + 1,$$

and we have by the Binomial Theorem that

$$\binom{m}{0} - \binom{m}{1} + \binom{m}{2} - \cdots + (-1)^{m}\binom{m}{m} = (1-1)^m = 0,$$

so the left-hand side of the equation equals 1, as desired.

3 Generalizing PIE for Counting Elements in More than 1 Set

Suppose we want to count the number of elements in at least k of n sets $A_1, A_2, \ldots, A_n$. The generalized form of PIE states that this count equals

$$\sum_{i=k}^{n} (-1)^{i-k} \binom{i-1}{k-1} S_i.$$

3.1 Understanding the Formula

To understand this formula, let's investigate an example. Say we have a group of students taking math, science, literature, and history classes. There are 60 students taking math and science; 52 taking math and literature; 49 taking math and history; 30 taking science and literature; 23 taking science and history; 21 taking literature and history; 25 taking math, science, and literature; 14 taking math, science, and history; 12 taking math, literature, and history; 11 taking science, literature, and history; and 9 taking all 4. How would we find the number of students taking at least 2 of the classes?

Starting without using the formula, we can begin writing an expression to count these students. We see that there are $60 + 52 + 49 + 30 + 23 + 21 = 235$ students taking 2 of the classes. However, students in 3 of the classes have been counted $\binom{3}{2} = 3$ times, as they are counted for each 2-element subset of the set of the 3 classes they are taking. We want to count each student exactly once, so we subtract 2 times the number of students in 3 of the classes, which is $25 + 14 + 12 + 11 = 62$, to get $235 - 2(62) = 111$. We have now overcounted the students taking all 4 classes; they have been counted $\binom{4}{2} - 2\binom{4}{3} = -2$ times, so we add back 3 times the number of students in all 4 classes, which is 9, to get $111 + 3(9) = 138$ students taking at least 2 of the classes.

Applying the formula directly yields a count of $\binom{1}{1}(235) - \binom{2}{1}(62) + \binom{3}{1}(9) = 138$ students in at least 2 of the classes, which matches our previous answer. But how do we know this formula always counts all elements correctly?

3.2 Proof of the Generalization of PIE

We must show that each element in at least k of the n sets $A_1, A_2, \ldots, A_n$ is counted exactly once. Consider some element x in r of the n sets, where $k \leq r \leq n$. Since x is counted $\binom{r}{i}$ times by S_i, we want to show

$$\binom{k-1}{k-1}\binom{r}{k} - \binom{k}{k-1}\binom{r}{k+1} + \binom{k+1}{k-1}\binom{r}{k+2} - \cdots + (-1)^{r-k}\binom{r-1}{k-1}\binom{r}{r} = 1.$$

We have omitted the subsequent terms because for $i > r$, $\binom{r}{i} = 0$ (alternatively, x is counted 0 times because it cannot be in the intersection of more than r sets). We now proceed by induction on k.

We have already proven the base case for $k = 1$; it is simply the formula from PIE for calculating $|A_1 \cup A_2 \cup \cdots \cup A_n|$. Then, suppose

$$\binom{j-1}{j-1}\binom{r}{j} - \binom{j}{j-1}\binom{r}{j+1} + \binom{j+1}{j-1}\binom{r}{j+2} - \cdots + (-1)^{r-j}\binom{r-1}{j-1}\binom{r}{r} = 1 \qquad (1)$$

for some $j \leq r \leq n$. We aim to show

$$\binom{j}{j}\binom{r}{j+1} - \binom{j+1}{j}\binom{r}{j+2} + \binom{j+2}{j}\binom{r}{j+3} - \cdots + (-1)^{r-j-1}\binom{r-1}{j}\binom{r}{r} = 1. \qquad (2)$$

By subtracting equation (2) from equation (1), we see that equation (2) holds if and only if

$$\binom{j-1}{j-1}\binom{r}{j} - \binom{j+1}{j}\binom{r}{j+1} + \binom{j+2}{j}\binom{r}{j+2} - \cdots + (-1)^{r-j}\binom{r}{j}\binom{r}{r} = 0,$$

using $\binom{a}{j-1} + \binom{a}{j} = \binom{a+1}{j}$ by Pascal's Identity for $j \le a \le n-1$. Note that

$$\begin{aligned}
&\binom{a}{b}\binom{b}{c} = \frac{a!}{b!(a-b)!} \cdot \frac{b!}{c!(b-c)!} = \frac{a!(a-c)!}{c!(a-c)!(a-b)!(b-c)!} = \binom{a}{c}\binom{a-c}{b-c} \\
\implies &\binom{j+s}{j}\binom{r}{j+s} = \binom{r}{j}\binom{r-j}{s},
\end{aligned}$$

which we can apply for $1 \le s \le r-j$ to get

$$\begin{aligned}
&\binom{j-1}{j-1}\binom{r}{j} - \binom{j+1}{j}\binom{r}{j+1} + \binom{j+2}{j}\binom{r}{j+2} - \cdots + (-1)^{r-j}\binom{r}{j}\binom{r}{r} \\
=&\binom{j-1}{j-1}\binom{r}{j} - \binom{r}{j}\binom{r-j}{1} + \binom{r}{j}\binom{r-j}{2} - \cdots + (-1)^{r-j}\binom{r}{j}\binom{r-j}{r-j} \\
=&\binom{r}{j}\left(\binom{j-1}{j-1} - \binom{r-j}{1} + \binom{r-j}{2} - \cdots + (-1)^{r-j}\binom{r-j}{r-j}\right) \\
=&\binom{r}{j}\left(\binom{r-j}{0} - \binom{r-j}{1} + \binom{r-j}{2} - \cdots + (-1)^{r-j}\binom{r-j}{r-j}\right) \\
=&\binom{r}{j}(0) \qquad \text{(Binomial Theorem)} \\
=&\,0,
\end{aligned}$$

as desired.

4 Conclusion

We have seen that the number of elements in at least k of n sets $A_1, A_2, \ldots, A_n$ is $\displaystyle\sum_{i=k}^{n}(-1)^{i-k}\binom{i-1}{k-1}S_i$. Notice that this sum takes the expression from PIE for finding $|A_1 \cup A_2 \cup \cdots \cup A_n|$ and multiplies each term by $\binom{i-1}{k-1}$ for $k \le i \le n$. It follows that the coefficient of these terms come from the kth diagonal of Pascal's Triangle, using 1-based indexing. This means that whenever we use PIE to count items with at least k of some n properties, we can find the coefficients by simply taking the terms of the kth diagonal of Pascal's triangle with alternating signs, starting with a positive coefficient.

References

Patrick, David. "A Piece of PIE." In *Intermediate Counting & Probability*, 49–82. Art of Problem Solving, 2007.